Completion of the omission on pairs

Human Compression project: a component of the L044 recovery

We acquire the complete proofs of *Centralizers of a primitive action on pairs*, *An alternating group on the pairs*, *Faithful symmetric lifts on pairs*, and *The central kernel in a symmetric action on pairs*. The last two are separate parts of the symmetric case; their full hypotheses and finite premises remain in force. We also acquire the complete finite maxima through degree fourteen and the retained L047 maximal-block decomposition. Each component has passed a distinct mathematical review. The complete proofs, rather than their result statements alone, are retained dependencies of this assembly.

Theorem 1. *Let $A_n = \langle \rho_0, \dots, \rho_{r-1} \rangle$ have a string C-marking in its natural action, where $n \geq 12$. Suppose some maximal omission is transitive and has a maximal block system consisting of pairs. Assume explicitly the alternating rank bound in every strictly smaller degree $12 \leq q < n$. Then $r \leq \lfloor (n-1)/2 \rfloor$.*

Proof. Write $n = 2m$. Degrees twelve and fourteen follow from the complete finite classification. We may therefore suppose $m \geq 8$ and, for a contradiction, $r \geq m$. The block image P is primitive, because the chosen block system is maximal. Choose an inclusion-minimal distinguished family L generating P, and let C be the other marks of the omission commuting with all of L, with R the remaining marks.

The complete centralizer component excludes the disconnected projected diagram: then $m \geq 60$ and $r \leq 2 \log_2 m + 6 < m$. In the connected case it establishes that L is an interval, $|C| \leq 1$, and $|R| \leq 1$. This last inequality uses the actual pair-centralizer theorem to make the omitted label a neighbour of L; it does not assume that this omitted permutation acts on the pairs. If $P \neq A_m, S_m$, the corrected primitive independent-string bound gives $|L| \leq m - 4$, hence $r = 1 + |L| + |C| + |R| \leq m - 1$, already a contradiction.

For $P = A_m$, the complete alternating-pair theorem gives the same bound. Its invocations of alternating rank use either the exact finite maxima or degrees $12 \leq q < 2m = n$, which are precisely the standing strong-induction hypotheses. For $P = S_m$, the faithful component first excludes $|L| = m - 1$ and every faithful projection of $\langle L \rangle$. It leaves exactly m even, $r = m$, $|L| = m - 2$, $C = \varnothing$, $|R| = 1$, with the central kernel $\langle z \rangle$ inside $\langle L \rangle$ and full even-weight kernel E inside the omission. After reversal the missing and extra marks lie at opposite ends of L. The complete central-kernel component excludes this exact configuration, including its independent degree-eight boundary. All primitive block images have therefore been exhausted. □

Attribution and scope. This assembles the three block-image cases of source Section 3.2. The source module reduction, L047 decomposition and acquired classifications retain their attribution; the actual centralizers, faithful endpoint arguments and string-only core argument are identified in the component proofs. No quotient IP or block action of a block-breaking generator is inferred. The theorem is a branch of the principal proof, not a whole-original recovery by itself.